

MEASURING WEIGHT

Weight is a measure of how heavy an object is. In the customary system, weight is measured in the three basic units shown in the table.

Unit	Sample Use	Comparing Units
ounce (oz)		$1\text{ oz} = \frac{1}{16}\text{ lb}$
pound (lb)		$1\text{ lb} = 16\text{ oz}$
ton (t)		$1\text{ t} = 2,000\text{ lb}$

Converting: Smaller ↔ Larger

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CONVERTING TO LARGER UNITS

Example Ibrahim bought a bag of candy weighing 45 oz. What is that weight in pounds and ounces?

Step 1

Set up the problem.

$$45 \text{ oz} = \underline{\quad} \text{ lb } \underline{\quad} \text{ oz}$$

The bag weighs **2 lb 13 oz**.

Step 2

Divide by the number of smaller units. (16 oz = 1 lb)

$$45 \text{ oz} \div 16 = 2 \text{ R}13$$

Step 3

Write the whole number as the larger unit and the remainder as the smaller unit.

$$45 \text{ oz} = 2 \text{ lb } 13 \text{ oz}$$

$$\begin{aligned} 1 \text{ lb} &= 16 \text{ oz} \\ 45 \div 16 &= 2 \text{ } ^{13}/_{16} \\ 2 \text{ lbs } 13 \text{ oz} \end{aligned}$$

$$\begin{array}{r} \text{scratch} \\ \hline 2 \\ 16 \overline{)45} \\ \underline{32} \\ 13 \end{array}$$

CONVERTING TO SMALLER UNITS

Example Heidi needed 60 oz of beef to serve her guests at a dinner party. She bought $3\frac{3}{4}$ lb of meat. Was that enough?

Step 1

Set up the problem.

$$3\frac{3}{4} \text{ lb} = \underline{\quad} \text{ oz}$$

Step 2

To change to a smaller unit, multiply by the number of smaller units.

$$3\frac{3}{4} \times 16 = \frac{15}{4} \times \frac{16^4}{1} = 60 \text{ oz}$$

Yes, Heidi has just enough meat (60 oz).

$$\begin{aligned} 40 \text{ oz} &= 3 \text{ lbs } 12 \text{ oz} \\ &= 3\frac{3}{4} \text{ lbs} \\ \therefore \text{ yes} \end{aligned}$$

$$\begin{array}{r} \text{scratch} \\ \hline 3 \\ 16 \overline{)60} \\ \underline{48} \\ 12 \end{array} \quad 12/16 = 3/4$$

Change each amount into larger or smaller units as indicated.

Change each amount into larger or smaller units as indicated.

1. 32 oz = 2 lb

16 oz = 1 lb

32 oz = 2 lb

92 oz = 5 lb 12 oz

$$\begin{array}{r} 5 \\ 16 \overline{) 92} \\ \underline{80} \\ 12 \end{array}$$

40 oz = 2 lb 8 oz

$$\begin{array}{r} 2 \\ 16 \overline{) 40} \\ \underline{32} \\ 8 \end{array}$$

4,500 lb = 2 t 500 lb

2000 lb = 1 ton

4000 lb = 2 ton

2. $1\frac{1}{2}$ lb = 24 oz

1 lb = 16 oz

1.5 lb = 1.5 (16)
= 24

$2\frac{1}{4}$ lb = 36 oz

1 lb = 16 oz

2.25 lb =
2.25 (16)
= 36

3 t = 6000 lb

1 ton = 2000 lb

3 tons =
3 (2000)
= 6000

$4\frac{3}{4}$ t = 9500 lb

4.75 (2000)
= 8000
+ 1500
= 9500

Adding & Subtracting Weight

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ADDING OR SUBTRACTING UNITS OF WEIGHT

Example Cobi roasted 2 chickens. The first weighed 4 lb 10 oz. The second weighed 3 lb 14 oz. What was the chickens' combined weight?

Step 1

Set up the problem.

$$\begin{array}{r} 4 \text{ lb } 10 \text{ oz} \\ + 3 \text{ lb } 14 \text{ oz} \\ \hline \end{array}$$

Step 2

Estimate first by adding the larger units.

$$4 \text{ lb} + 3 \text{ lb} = 7 \text{ lb}$$

The answer should be slightly more than 7 lb.

Step 3

Add or subtract the units separately.

$$\begin{array}{r} 4 \text{ lb } 10 \text{ oz} \\ + 3 \text{ lb } 14 \text{ oz} \\ \hline 7 \text{ lb } 24 \text{ oz} \end{array}$$

Step 4

Regroup or simplify the answer, if necessary.

$$\begin{array}{r} 7 \text{ lb } 24 \text{ oz} = \\ 7 \text{ lb} + 1 \text{ lb } 8 \text{ oz} = \\ 8 \text{ lb } 8 \text{ oz} \end{array}$$

8 lb 8 oz is slightly more than 7 lb. The combined weight is **8 lb 8 oz**.

$$\begin{array}{r} 4 \text{ lb } 10 \text{ oz} \\ + 3 \text{ lb } 14 \text{ oz} \\ \hline 7 \text{ lb } 24 \text{ oz} \\ = 8 \text{ lb } 8 \text{ oz} \end{array}$$

$$\begin{array}{r} 1 \\ 16 \overline{) 24} \\ \underline{16} \\ 8 \end{array}$$

Add or subtract the following weights.

3. $\begin{array}{r} 2 \text{ lb } 3 \text{ oz} \\ + 1 \text{ lb } 8 \text{ oz} \\ \hline \end{array}$

$$3 \text{ lb } 11 \text{ oz}$$

$$\begin{array}{r} 6 \text{ lb } 14 \text{ oz} \\ + 4 \text{ lb } 11 \text{ oz} \\ \hline \end{array}$$

$$\begin{array}{r} 10 \text{ lb } 25 \text{ oz} \\ = 11 \text{ lb } 9 \text{ oz} \end{array}$$

$$\begin{array}{r} 2 \text{ t } 152 \text{ lb} \\ + 1 \text{ t } 576 \text{ lb} \\ \hline \end{array}$$

$$3 \text{ tons } 728 \text{ lb}$$

$$\begin{array}{r} 3 \text{ t } 1,500 \text{ lb} \\ + 5 \text{ t } 1,250 \text{ lb} \\ \hline \end{array}$$

$$\begin{array}{r} 8 \text{ tons } 2750 \text{ lb} \\ = 9 \text{ tons } 750 \text{ lb} \end{array}$$

4. $\begin{array}{r} 5 \text{ lb } 12 \text{ oz} \\ - 3 \text{ lb } 8 \text{ oz} \\ \hline \end{array}$

$$2 \text{ lb } 4 \text{ oz}$$

$$\begin{array}{r} 7 \text{ lb } 3 \text{ oz} \\ - 5 \text{ lb } 12 \text{ oz} \\ \hline \end{array}$$

$$\begin{array}{r} = 6 \text{ lb } 19 \text{ oz} \\ - 5 \text{ lb } 12 \text{ oz} \\ \hline 1 \text{ lb } 7 \text{ oz} \end{array}$$

$$\begin{array}{r} 12 \text{ lb } 14 \text{ oz} \\ - 6 \text{ lb } 15 \text{ oz} \\ \hline \end{array}$$

$$\begin{array}{r} = 11 \text{ lb } 30 \text{ oz} \\ - 6 \text{ lb } 15 \text{ oz} \\ \hline 5 \text{ lb } 15 \text{ oz} \end{array}$$

$$\begin{array}{r} 4 \text{ t } 800 \text{ lb} \\ - 1 \text{ t } 1,200 \text{ lb} \\ \hline \end{array}$$

$$\begin{array}{r} = 3 \text{ t } 2800 \text{ lb} \\ - 1 \text{ t } 1200 \text{ lb} \\ \hline 2 \text{ t } 1600 \text{ lb} \end{array}$$

Applications

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Solve the following word problems.

5. Trevor used about half of a 5-lb bag of potatoes (2 lb 6 oz). How much did the remaining potatoes weigh?

$$\begin{array}{r} 5 \text{ lbs} \\ - 2 \text{ lbs } 6 \text{ oz} \\ \hline 4 \text{ lbs } 16 \text{ oz} \\ - 2 \text{ lbs } 6 \text{ oz} \\ \hline 2 \text{ lbs } 10 \text{ oz} \end{array}$$

6. Tyus can carry $1\frac{1}{2}$ t of cargo in his truck. He loads the truck with 1,756 lb of bricks. How many more pounds of bricks can Tyus put in his truck?

$$\begin{array}{r} 1\frac{1}{2} \text{ tons} \\ - 1756 \text{ lbs} \\ \hline = 3000 \text{ lbs} \\ - 1756 \text{ lbs} \\ \hline 1244 \text{ lbs} \end{array}$$

7. McGarity's Bar and Grill had 35 lb 3 oz of steak left over from last night. They received a delivery of 14 lb 15 oz today. How much steak do they have altogether?

$$\begin{array}{r} 35 \text{ lbs } 3 \text{ oz} \\ + 14 \text{ lbs } 15 \text{ oz} \\ \hline 49 \text{ lbs } 18 \text{ oz} \\ = 50 \text{ lbs } 2 \text{ oz} \end{array}$$

8. Keiko is using a forklift to load shipping crates on a freight elevator. Each crate weighs 220 lb. The elevator can hold 1 t safely. Can she safely load 9 crates?

$$(9 \text{ crates})(220 \text{ lbs}) = 1980 \text{ lbs}$$

∴ yes!!

$$\begin{array}{r} \text{scratch} \\ 220 \\ \times 9 \\ \hline 1980 \end{array}$$

9. **Discuss** Which customary weight units would you use to measure a truck? A man? A box of cereal?

truck : tons

man : pounds

cereal : ounces